THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE TWO-CHAPTER TEST SET C

Chapters 1 & 2: **Exploration: Entering the World of Chemistry, Exploring Mixtures and Their Separation** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. In a simple physics model, a moving car is often represented as: 1

(a) a sphere with bonds **(b)** a cell diagram
(c) a layered sphere **(d)** a single point

2. The speed of light is now defined to be exactly: 1

(a) 300 m/s **(b)** 299792458 m/s **(c)** 186000 m/s only **(d)** 3 km/s

3. In science, a ‘theory’: 1

(a) means an untested guess
(b) cannot change with new evidence
(c) provides a tested explanation of why patterns occur
(d) is the same as a law

4. A well-tested scientific theory: 1

(a) is rejected if a student disagrees
(b) is final and can never be questioned
(c) may be improved or revised when new evidence appears
(d) is just an opinion

5. A claim says ‘food becomes harmful during an eclipse’. The best scientific response is to: 1

(a) ask what mechanism and measurable change could make it harmful
(b) accept it because it is old
(c) accept it because many people share it
(d) reject it without asking anything

6. 12 g of salt is dissolved in 88 g of water. The mass by mass percentage of the solution is: 1

(a) 12 % **(b)** 6 % **(c)** 13.64 % **(d)** 22 %

7. 18 g of a solute is dissolved in water to make 200 mL of solution. Its mass by volume percentage (% m/v) is: 1

(a) 8.26 % **(b)** 9 % **(c)** 4.5 % **(d)** 18 %

8. The solubility of gases in liquids generally: 1

(a) increases with temperature
(b) decreases with an increase in temperature
(c) stays the same
(d) is zero at room temperature

9. Which method separates two miscible liquids with boiling points differing by more than about 25 °C? 1

(a) filtration **(b)** simple distillation
(c) separating funnel **(d)** sublimation

10. A mixture of camphor and sand can be separated by: 1

(a) crystallization **(b)** distillation

(c) sublimation **(d)** separating funnel

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. Estimate the number of breaths a person takes in a day if the rate at rest is 14 breaths per minute. 2

12. Name two branches of science involved in studying a pressure cooker and one idea from each. 2

13. State two differences between a true solution and a suspension. 2

14. What is the Tyndall effect? Give one example. 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 30–40 words each.

15. What is the role of repeated observations and measurement refinement in building scientific knowledge? 3

16. Explain with two examples why the right proportion of solute to solvent is important in daily life. 3

17. Give two examples where predictions from science help in daily life. 3

OR

A student's prediction about the result of a match was based on lucky colour. Explain why this is not a scientific prediction. 3

18. Why do carbonated drinks lose their fizz when kept warm? Relate to solubility. 3

OR

What are alloys? Why are they useful? Give two examples. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study:** A school group is told ‘social media forwards are always right’. 5

(i) How should they check a claim?

(ii) Give an example claim and the question that disproves it.

(iii) What habits does secondary science build?

(iv) Why do scientists welcome their ideas being tested?

OR

(i) Explain, with two examples, why real-world problems need several branches of science. 5

(ii) Draw a small mind-map describing how a mobile phone involves at least two branches.

(iii) What is the benefit of connecting science with mathematics, technology and arts?

20. (i) Why is the solubility of gases in liquids important for aquatic life? 5

(ii) What happens to dissolved oxygen when water becomes warmer?

(iii) Explain using solubility curves.

OR

Case study: ORS is prepared with specific amounts of salt and sugar in water. 5

(i) Why is changing the amounts harmful?

(ii) 20 g sugar and 3.5 g salt are in 1 L; find % m/v of each.

(iii) Why is mass by volume % used in medicines?

END OF QUESTION PAPER Total: 40 Marks